

torch.eye#

<https://pytorch.org/docs/stable/generated/torch.eye.html#torch.eye>

Description:

Returns a 2-D tensor with ones on the diagonal and zeros elsewhere. `n` (int) – the number of rows `m` (int, optional) – the number of columns with default being `n` `out` (Tensor, optional) – the output tensor. `dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`).

Function Signature

```
torch.eye(n, m=None, *, out=None, dtype=None,
layout=torch.strided, device=None, requires_grad=False) →
Tensor#
```

Parameters

Keyword Arguments

Returns

Return type

Parameters

Keyword

out (Tensor, optional) – the output tensor. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

Returns:

Tensor

Code Examples

```
>>> torch.eye(3)
tensor([[ 1.,  0.,  0.],
        [ 0.,  1.,  0.],
        [ 0.,  0.,  1.]])
```

Detailed Documentation

Documentation

Returns a 2-D tensor with ones on the diagonal and zeros elsewhere.

`n` (int) – the number of rows

`m` (int, optional) – the number of columns with default being `n`

`out` (Tensor, optional) – the output tensor.

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`).

`layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`.

`device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

`requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

A 2-D tensor with ones on the diagonal and zeros elsewhere